



Electric field sculpting in ultra-low noise HgCdTe avalanche photodiodes

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ABSTRACT

In advanced linear-mode infrared avalanche photodiodes (APDs), microscopic doping distributions and geometric dimensions fundamentally dictate avalanche gain and excess noise. This research investigates the internal space-charge dynamics of Mercury Cadmium Telluride (HgCdTe) APDs, introducing the design strategy of Electric Field Sculpting to elucidate non-equilibrium carrier transport. Employing a rigorous microscopic-mesoscopic Monte Carlo framework, this study quantifies the impact of doping differentials across independent functional zones (p, i, n +) alongside the geometric width of the multiplication zone (i-region). Unregulated doping discrepancies sculpt precipitous local field gradients, violently ejecting electrons into low-field zones post-ionization. This induces catastrophic polar optical phonon (POP) emission, shattering the deterministic periodicity of the dead-space effect and drastically inflating the macroscopic excess noise factor. Conversely, a smoothly sculpted electrostatic gradient tightly suppresses statistical gain variance. Furthermore, passively extending the multiplication width beyond maximum depletion limits creates an electrically neutral dead zone. This geometric miscalibration introduces a diffusion tail, severely bottlenecking high-frequency bandwidth. By systematically evaluating multidimensional physical trade-offs—including temporal dynamic response, tunneling dark current, and noise factor—this research architectures an optimally sculpted structure characterized by full depletion and gentle electrostatic gradients. These insights establish a robust theoretical foundation for designing next-generation, ultra-low-noise, high-bandwidth infrared photonic sensors.

1. Introduction

The relentless demand for next-generation photonic technologies—spanning deep-space optical communication, high-resolution LiDAR, and ultra-faint signal detection—requires photodetectors with single-photon sensitivity, sub-nanosecond resolution, and ultra-low dark currents [1]. Operating across a broad spectral range encompassing the short-wave infrared (SWIR) to the long-wave infrared (LWIR), the ternary alloy Mercury Cadmium Telluride (HgCdTe) remains the paramount material system for advanced linear-mode avalanche photodiodes (LmAPDs) [2–6]. In particular, Hu et al. proposed a ingenious approach of back-to-back PN junctions for separating dual-band

photogenerated carriers, and the spatially and temporally highly synergistic long-wavelength/mid-wavelength HgCdTe infrared detector began to exhibit significant advantages, effectively suppressing spectral cross-talk achieving performance levels comparable to international state-of-the-art standards [7]. Furthermore, the integration of surface plasmon polaritons (SPPs) and surface phonon polaritons (SPhPs) into infrared detectors, which simultaneously enhances quantum efficiency, bandwidth, and multi-dimensional detection capability, has also become an important future development trend [8]. Recently, an appealing platform for intelligent detections based on ultra-high ferroelectric localized fields through a reconfigurable atomic layer structure has been proposed, and thus the recognition ability of the integrated

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architecture can be significantly enhanced [9]. Its unique bandgap architecture strongly suppresses hole-initiated impact ionization, enabling a near-ideal “pure electron” avalanche multiplication. This deterministic mechanism generates exponential avalanche gain while maintaining an excess noise factor (ENF) that approaches the absolute theoretical minimum [10–13].

However, as photodiode devices evolve towards smaller feature sizes, the thickness of the internal multiplication region (i-region) is continuously compressed, and the electric field intensity sharply increases. In the narrow high-field depletion region, carrier transport is no longer in thermodynamic equilibrium, and the continuous multiplication model gradually fails, being replaced by the “dead-space effect” dominated by non-equilibrium carriers [14]. Although theoretically supporting extremely low noise detection, in the actual device fabrication, different devices often exhibit significant performance differentiation. As shown in Fig. 1(a), within the critical operating bias range of 3 V to 5 V, the classic devices reported in the literature (such as Rothman 2009) [15] exhibit more abnormal ultra-high excess noise compared to the benchmark experimental devices. Meanwhile, the gain and noise rise curves in Fig. 1(b) indicate that there is a significant lateral drift in the turn-on voltage and curve slope among different devices. After analysis, it was found that the root cause of this macroscopic performance degradation lies in the differences in doping concentration levels in specific regions within the device and the variation in the geometric width of the i-region, which leads to a redistribution of the space charge density and significantly reshapes the internal electrostatic field profile [16]. This distortion of the electrostatic field profile severely modulates the microscopic scattering mechanism of carriers, causing a sharp expansion of the spatial variance of the dead space and ultimately resulting in significant fluctuations in macroscopic gain. To uncover the mechanism behind this phenomenon, this study employed a micro-mesoscopic Monte Carlo model [14,17] to comprehensively quantify the impact of structural parameter changes on the electric field and microscopic dynamics and thereby demonstrated an optimal structure that combines high dynamic bandwidth with extremely low noise.

2. Electrostatic sculpting via epitaxial doping gradients

When analyzing the complex carrier dynamics within sub-micron APDs, traditional macroscopic phenomenological models (such as drift–diffusion equations) fail because they erroneously presume charge carriers exist in a state of local thermodynamic equilibrium. This assumption obscures the non-local impact ionization history, velocity overshoot, and ballistic transport characteristics inherent to intense electric fields. To rigorously capture the spatial modulation dictated by the dead-space effect, this study employs a micro-mesoscopic Monte Carlo (MC) model to solve the highly non-linear Boltzmann Transport Equation (BTE) from first principles. By tracking the high-resolution, step-by-step evolution of a vast ensemble of discrete initial electrons, the MC model generates spatially resolved impact ionization statistics [18]. These detailed transport histories provide definitive physical evidence for dissecting the root causes of macroscopic performance fluctuations. Simulations were conducted at 77 K. As supported by recent characterizations [19], minor temperature fluctuations in this cryogenic regime have a negligible impact on device performance and the stability of the sculpted electrostatic gradient.

To identify the optimal device parameters, this research rigorously isolated two core variables, “doping concentration differentials” and “i-region physical width”, to extract their reshaping rules on the one-dimensional electrostatic field. According to the classical Poisson equation and weak oscillation theory [14], the net uncompensated space-charge density fundamentally defines both the curvature and the spatial decay slope of the localized electric field. This relationship is mathematically expressed as $\nabla^2\phi = -\rho/\epsilon_s$, which simplifies to $dE/dx = qN_{\text{eff}}(x)/\epsilon_s$ in a one-dimensional case. This expression indicates that the local gradient of the internal electric field is directly proportional to the net effective doping concentration, where ϕ is the electrostatic potential, ρ is the local space-charge density, ϵ_s is the static dielectric permittivity of the semiconductor material, E represents the internal electric field magnitude, x is the spatial coordinate (depth), q is the elementary charge, and $N_{\text{eff}}(x)$ denotes the localized net effective doping concentration profile. By statistically analyzing the non-equilibrium transport

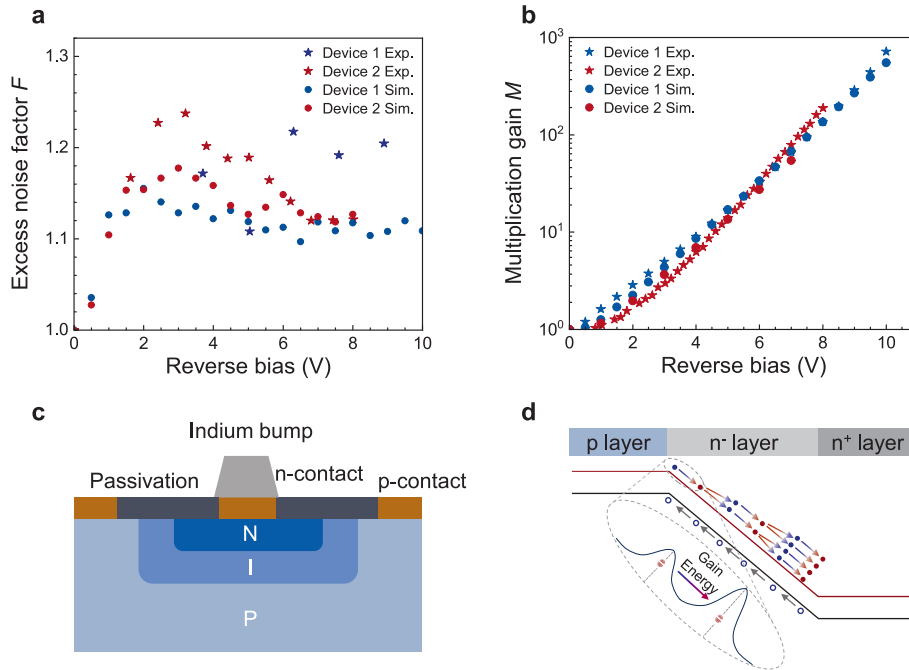


Fig. 1. Benchmark device performance and schematic structure. (a) Excess noise factor versus reverse bias, comparing the benchmark experimental device (Device 1) with data from Rothman 2009 (Device 2); (b) Avalanche gain versus reverse bias for the benchmark experimental device (Device 1) and the Rothman 2009 device (Device 2); (c) Cross-sectional schematic diagram of the planar PIN APD structure; (d) Meso-physical images of electron avalanche propagation and microscopic trajectories within the multiplication zone.

history of a large number of initial electrons, the microscopic-mesoscopic Monte Carlo model outputs highly spatially resolved statistical charts of ionization events, providing solid physical evidence for dissecting macroscopic performance fluctuations [20]. Based on the benchmark device and the device parameters optimized by Rothman in 2009, the control logic of the macroscopic noise caused by the doping concentration variation in different regions is accurately isolated. The fundamental reason for the macroscopic performance differences between different devices in Fig. 1(a) and (b) is the change in doping concentration at adjacent regions (such as the p-region and the i-region). The numerical simulation employs composition-(x) and temperature-(T) dependent empirical models to define the essential material parameters. The bandgap energy (E_g , in eV) is calculated as $E_g = -0.302 + 1.93x - 0.810x^2 + 0.832x^3 + 5.35 \times 10^{-4}(1 - 2x)T$ [21]. The electron effective mass (m_e^*) is coupled to the local bandgap via $m_e^*/m_0 = [-0.6 + 6.333(2/E_g + 1/(E_g + 1))]^{-1}$, [22] while the heavy-hole effective mass is fixed at $m_h^* = 0.55m_0$. Consequently, the effective densities of states (N_c and N_v) scale proportionally with $T^{3/2}$. Carrier transport is governed by a temperature-dependent electron mobility (μ_n), with the hole mobility defined as $\mu_p = 0.01\mu_n$. Additionally, the static dielectric constant is $\epsilon_r = 11.8$, and the electron affinity (χ , in eV) linearly scales with E_g as $\chi = 4.23 - 0.813(E_g - 0.083)$ [4]. According to Poisson's equation, when the background doping concentration changes, the net space charge density at the junction boundary also

changes. The spatial decay slope of the internal electric field (i.e., the first derivative) is proportional to the doping concentration of the region. Therefore, as shown in Fig. 2(d), the geometric splicing of different doping concentrations leads to completely different electric field profiles. When the charge density on one side of the junction is higher, a sharp electric field peak forms at the interface between the p-region and the i-region and rapidly decays with a large slope into the i-region. This electric field distortion caused by doping concentration differences makes the carriers face an extremely unstable electrostatic driving force during the microscopic transport process, which is the main decisive factor leading to abnormal macroscopic noise.

As shown in Fig. 2(a), when the doping concentration difference between regions is significant, a steep electric field with a large slope is excited (see the extracted curve corresponding to Fig. 2(d)). The pioneering theory of localized photonic-electronic co-modulation [23] demonstrated that the manipulation of photogenerated carriers via localized electric fields can achieve dark current suppression by more than an order of magnitude. A consistent phenomenon is observed in our work: the photogenerated electrons are instantaneously accelerated by a very high electric field when crossing the junction, and the first-generation ionization occurs very quickly (corresponding to the extremely narrow first-generation ionization peak in Fig. 3(h)). However, this non-uniform distribution introduces severe carrier transport limitations. Due to the rapid drop in the electric field, the efficiency of electrons obtaining kinetic energy from the electrostatic field

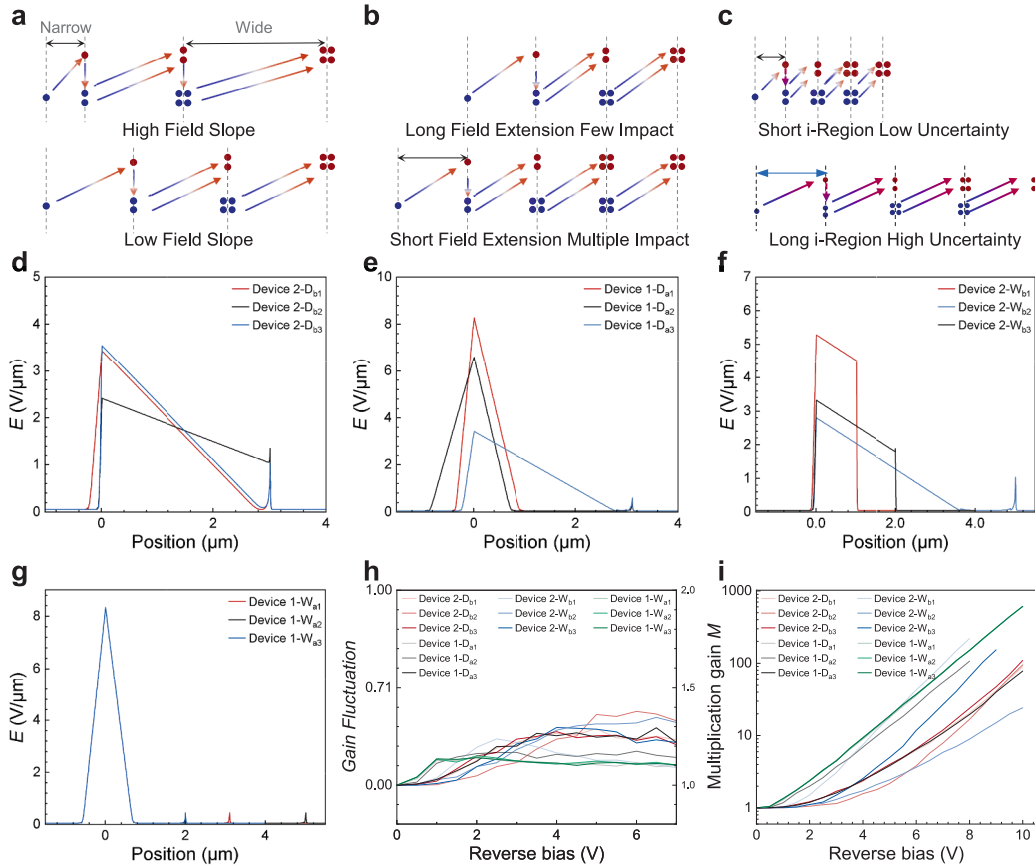


Fig. 2. Electrostatic field reshaping and its impact on excess noise and gain. (a) Schematic diagram of the effects of steep and gentle electric field gradients on carrier impact ionization; (b) Schematic diagram of the effect of electric field extension depth (effective avalanche volume) on carrier impact ionization; (c) Schematic diagram of the influence of electric field peak separation on carrier impact ionization caused by the increase of the width of the i zone; One-dimensional electric field spatial distributions corresponding to different structural parameters: (d) Device 2 (D_b series) and (e) Device 1 (D_a series) under varying doping levels; (f) Device 2 (W_b series) under varying i-region widths; and (g) Device 1 (W_a series) illustrating the incomplete depletion state caused by excessive i-region width. (h) Gain fluctuation (excess noise factor) and (i) avalanche multiplication gain M as a function of reverse bias for all device configurations. (Note: The W series in the legends represent different i-region widths, while the D series represent different doping levels.).

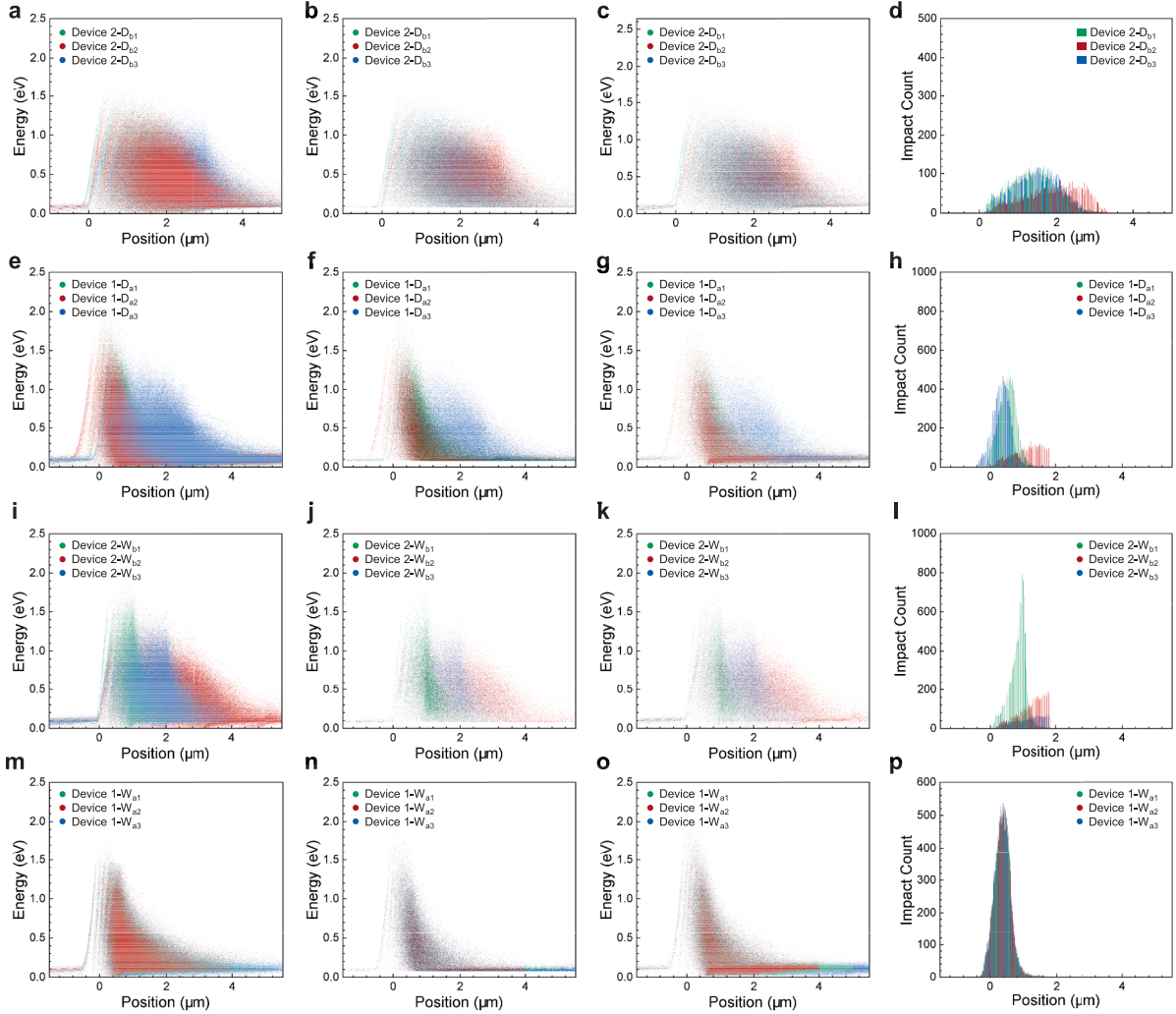


Fig. 3. Monte Carlo microscopic statistics of electron dynamics. The columns from left to right illustrate the spatial evolution of electron energy (1st column), polar optical phonon (POP) emission (2nd column), POP absorption (3rd column), and the spatial distribution of impact ionization counts (4th column, spatial bin size: 10 nm). (a)-(d) Device 2 (D_b series): Microscopic states under a rapidly dropping electric field, showing dispersed ionization peaks due to severe energy trajectory collapse. (e)-(h) Benchmark experimental device, Device 1 (D_a series): Microscopic states under an optimized smooth linear electric field, demonstrating highly localized, periodic ionization peaks and minimal physical broadening. (i)-(l) Device 2 (W_b series): Effects of artificially increasing the geometric width of the i-zone, resulting in a significantly delayed initial ionization onset position ($\sim 1.0 \mu\text{m}$) due to enhanced phonon dissipation. (m)-(p) Device 1 (W_a series): Dynamics in a non-complete depletion state, where the avalanche process stalls completely after a single primary ionization peak due to the lack of an effective driving electric field in the quasi-neutral region.

significantly decreases after entering the low-field region, and the energy evolution trajectory collapses severely (see Fig. 3(a)). Under insufficient electrostatic acceleration force, electrons have difficulty crossing the resonant region of polar optical phonon (POP) scattering cross-sections, leading to an exponential and explosive increase in the POP emission probability (see Fig. 3(b)). Quantitatively, data extracted from our microscopic database reveals that the optimally sculpted field provides sustained acceleration to overcome the massive POP emission rate ($10^{12} - 10^{13} \text{ s}^{-1}$) that otherwise traps electrons and erases spatial memory under steep gradients. This mechanism is explicitly verified by the scatter distributions: under the steep field (Fig. 3b), carrier energies are heavily dissipated and trapped between 0.1 eV and 1.0 eV, causing POP emission events to uncontrollably disperse across a broad spatial range from 0.5 μm to 4.0 μm . Conversely, the sculpted field (Fig. 3f) enables carriers to reach higher pre-emission energies ($\sim 1.3 \text{ eV}$) and tightly confines the emission space, terminating sharply near 2.5 μm . Furthermore, artificially widening the i-zone (Fig. 3j) severely delays the onset of dense emission to $\sim 1.0 \mu\text{m}$, while incomplete depletion (Fig. 3n and o) causes energies to collapse into a flat $\sim 0.1 \text{ eV}$ threshold band.

Electrons are forced to dissipate kinetic energy by continuously emitting phonons, resulting in significant uncertainty in the physical location of subsequent impact ionizations. With the help of the spatial avalanche theory that is in high agreement with experimental results [14], this is reflected in the ionization count histogram (Fig. 3(d)), where the subsequent generation ionization peaks rapidly disperse and overlap, spanning a wide spatial range from 0 to 4 μm with peak counts struggling to exceed 150, completely weakening the periodic characteristics of the dead zone effect. Macroscopically, this is manifested as the loss of control of gain fluctuations and a sharp increase in the excess noise factor (as shown in Fig. 2(h) and (i)). Correspondingly, by finely adjusting the doping level to reduce the concentration difference, the smooth linear electric field decay slope shown in Fig. 2(a) and (d) can be calculated. When electrons traverse the multiplication region, the electrostatic acceleration force they experience remains in a relatively constant and sufficient smooth range. Therefore, each recovery process after energy reset has high temporal and spatial predictability. At the microscopic level, Fig. 3(h) clearly shows that the physical broadening of each generation of ionization peaks is extremely small, and the impact

events are uniformly and orderly distributed in three-dimensional space. The primary ionization peak is sharply localized at $\sim 0.5 \mu\text{m}$, followed by highly distinct subsequent generation peaks at $\sim 0.9 \mu\text{m}$ and $\sim 1.3 \mu\text{m}$, maintaining concentrated peak counts of over 400.

This smooth step structure fully exploits the dependence of the dead zone effect on the memory of the carrier transport history, suppressing the statistical variance of a single avalanche process to the theoretical limit and maintaining the excess noise at an extremely low level.

In addition to regulating the local slope, as shown in Fig. 2(b) and (e), the doping configuration of each region also determines the penetration depth of the strong electric field into the depth from the Poisson boundary condition, that is, physically defining the effective “avalanche volume” of the device. For the “short effective depletion extension” structure, the spatial span of the high-field region is greatly compressed. Before the primary electrons leave the high-field region, they can only complete a few dead zone cycles, which severely restricts the macroscopic avalanche multiplication limit that the device can achieve under a safe bias, making it difficult to meet the high gain requirements for long-distance weak target detection. However, in the optimized “deep penetration depletion region” structure, the strong electric field effectively extends deep into the depletion region. The microscopic statistics in Fig. 3(a) confirm that the electron ensemble can maintain a high-energy excited state over a wide physical span, and the POP phonon emission (Fig. 3(b)) transforms into a relatively uniform low-impact background. The long effective extension structure (Fig. 3(d)) clearly shows more than five consecutive, independent, and distinct ionization generation peaks, which from a microscopic dynamic perspective, firmly indicates that electrons have achieved highly controllable continuous reproduction within the effective electric field, enabling the device to stably output high gain under a moderate bias.

Under the premise of a fixed bias and doping, simply increasing the physical width of the i-region will introduce significant negative effects. As shown in Fig. 2(c), artificially increasing the physical width of the i-region will directly reduce the internal potential gradient, resulting in a clear separation and a decrease in the absolute value of the peak electric field in space (see Fig. 2(f)). Under a low background electric field level, the rate of electron kinetic energy acquisition is severely hindered (Fig. 3(i)), forcing them to have a longer time window to interact strongly with lattice polar phonons (Fig. 3(j)). This strong phonon dissipation reduces the spatial certainty of ionization dominated by the dead zone effect. As shown in Fig. 3(l), the spatial onset of the first ionization peak is significantly delayed to approximately $1.0 \mu\text{m}$, leading to the gain of the device being extremely difficult to start under low voltage and accompanied by high noise variance. Another situation, as shown in Fig. 2(g), when the width of the i-region exceeds the physical limit of the maximum depletion layer under the corresponding bias, the device will not be completely depleted. Currently, the effective high-field region space is completely frozen, and the edge non-depleted region is completely degraded into a quasi-neutral region lacking an electric field drive. Due to the local electric field $E(x)$ in the internal core avalanche region ceasing to evolve, the microscopic statistics in Fig. 3(m)–(o) show that the acceleration trajectory and phonon emission and absorption characteristics of electrons within the effective region present a meaningless constant state lacking modulation, with electron energies capped near 1.5 eV without further cyclic evolution. In the redundant quasi-neutral region, carriers lose the electrostatic field drift force and can only rely on the extremely slow concentration gradient for diffusion transport. This avalanche failure is starkly evident in Fig. 3(p), where only a single, massive initial ionization peak occurs at $\sim 0.5 \mu\text{m}$ (exceeding 500 counts), with the complete absence of subsequent multiplication generations. This diffusion mechanism severely hinders the transient collection process of photoelectrons, leading to a significant degradation in macroscopic dynamic response performance.

3. Electrostatic sculpting via epitaxial doping gradients

Based on the established physical rules, simplistic optimization strategies—such as blindly maximizing the peak electric field or passively widening the geometry—inevitably lead to comprehensive performance collapse [24,25]. The architectural design of a state-of-the-art HgCdTe APD must navigate a highly complex, Multiphysics coupled equilibrium between doping gradients, depletion limits, and geometric dimensions. To achieve this complex multiphysics equilibrium, the epitaxial structure was meticulously refined. The detailed physical parameters of the optimally sculpted PIN architecture, including the precise geometric widths, material compositions, and doping gradients for each distinct region, are quantitatively summarized in Table 1. By replacing abrupt doping steps with a gently sculpted electrostatic profile, this architecture significantly relaxes the extreme manufacturing precision typically required during epitaxial growth, offering higher tolerance to inevitable local doping fluctuations. Practically, the $5 \times 10^{15} \text{ cm}^{-3}$ background doping is achievable via modern Molecular Beam Epitaxy (MBE) [12,26]. Furthermore, this sculpted field exhibits inherent tolerance to minor epitaxial doping fluctuations by preventing localized ultra-high field spikes and tunneling.

The time-domain dynamic response speed, quantified by the 3 dB cutoff frequency (f_{3dB}), dictates the absolute data throughput limit for spatial LiDAR mapping and free-space optical communications [27]. In poorly sculpted, incompletely depleted structures, the carrier transit time is dominated by slow diffusion ($t_{diff} = \frac{W^2}{2D}$), leading to a massive diffusion tail in the arrival time spectrum. This severely increases the response time constant, causing critical low-frequency attenuation. The optimally architected structure precisely aligns the physical epitaxial thickness of the i-region with its specific maximum depletion limit, ensuring a strictly “fully depleted” scenario. By completely excising the quasi-neutral zone, carriers remain permanently locked in a state of high-electric-field drift at saturated velocity (v_{sat}). This severs the diffusion tail, yielding an astonishingly sharp pulse response that propels the bandwidth deep into the multi-gigahertz (GHz) echelon:

$$f_{3dB} \approx \frac{0.44}{t_{tr}} = \frac{0.44 \cdot v_{sat}}{W}$$

where t_{tr} represents the carrier transit time, v_{sat} is the saturated drift velocity, and W is the effective width of the fully depleted multiplication region. As demonstrated in Fig. 4(b), the optimal structure exhibits a highly concentrated time-domain response peak arriving rapidly at $\sim 100 \text{ ps}$, without the persistent slow-diffusion tail observed in the non-optimal variants. Consequently, this translates to a superior frequency-domain performance in Fig. 4(a), where the optimal design propels the 3 dB bandwidth deep into the multi-gigahertz echelon, achieving a cutoff frequency exceeding 7 GHz, which is notably higher than the comparative structures. It should be noted that this multi-GHz bandwidth reflects the intrinsic transit-time limit of the device; practical system-level RC time constants induced by specific hybridization and ROIC architectures are not considered in these Monte Carlo results. Furthermore, the optimal structure delivers peak Signal-to-Noise Ratio (SNR) characteristics. By seamlessly sculpting the electric field into a gentle slope, the microscopic gain fluctuation variance is violently suppressed. The macroscopic excess noise factor ($F(M)$) remains tightly bound to its fundamental limit even under elevated reverse biases, directly maximizing system sensitivity:

Table 1
Structural parameters of the optimized PIN APD.

Region	Thickness(μm)	Doping density(at/cm ³)	Composition x in Cd
n+	2	2×10^{17}	0.3
n-	2	5×10^{15}	0.3
p	4.5	8×10^{16}	0.3

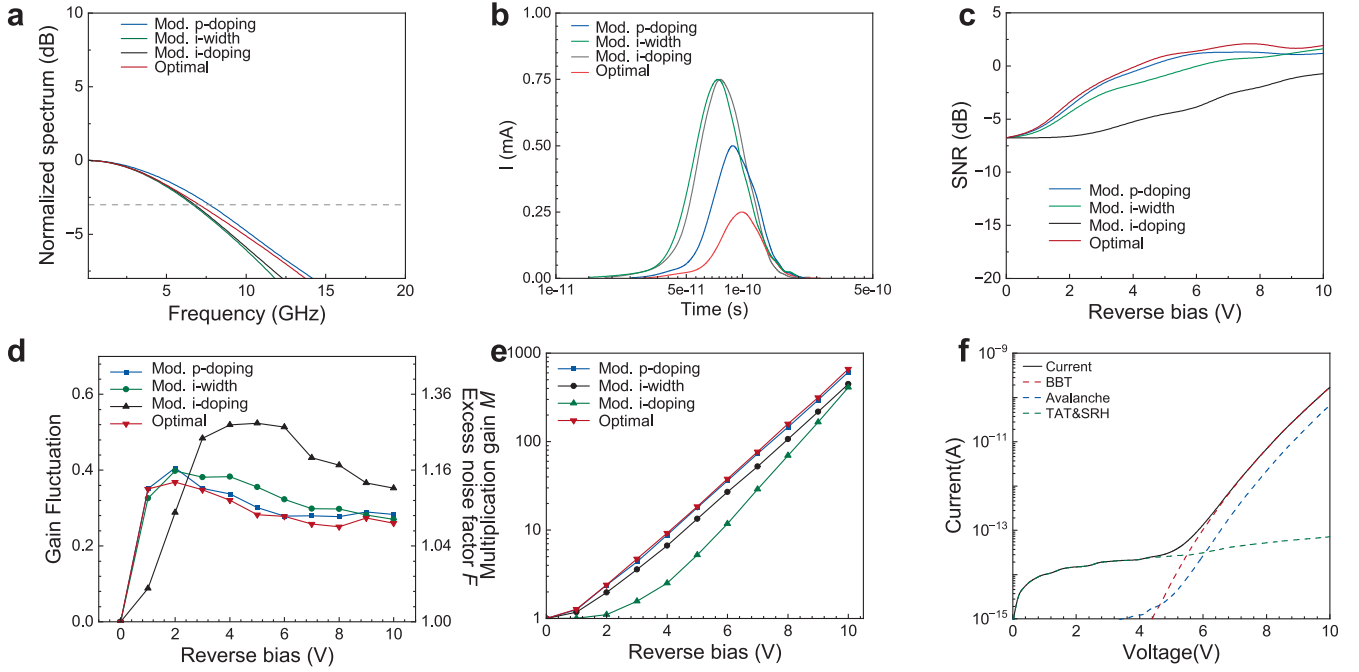


Fig. 4. Comprehensive evaluation of multidimensional physical quantities for the optimal structure and its variants. (a) Normalized frequency-domain response spectra. (b) Time-domain distribution of electron and hole arrival times (impulse response). (c) Evaluation of Signal-to-Noise Ratio (SNR) levels for different structures under varying biases. (d) Gain fluctuation (excess noise factor) versus operating bias. (e) Avalanche multiplication gain versus operating bias. (f) Macroscopic dark current (I-V) characteristics of the optimal structure, showing the breakdown of different current mechanisms (BBT, Avalanche, TAT & SRH). *Optimal* refers to the fully depleted baseline design with a sculpted electric field; *Mod. p-Doping*, *Mod. i-Width*, and *Mod. i-Doping* represent comparative structures with modified p-region doping, i-region width, and i-region doping based on the optimal design, respectively.

$$SNR = \frac{(MI_{ph})^2}{2q(I_{ph} + I_d)M^2F(M)\Delta f}$$

where M denotes the avalanche gain, I_{ph} and I_d are the primary (unmultiplied) photocurrent and dark current respectively, q is the elementary charge, $F(M)$ is the excess noise factor, and Δf is the operational measurement bandwidth. Crucially, this intelligently sculpted landscape inherently restricts the peak electric field below the critical threshold that triggers severe band-to-band tunneling (BBT) and trap-assisted tunneling (TAT). In Fig. 4f, we adopted the TAT model, [28] which currently shows the best agreement with experiments,

$$I_{stat} = -A \cdot \frac{\pi^2 q^2 N_i m_e^* M^2 (V_{bi} - V_d) \exp\left(-\frac{\sqrt{3} E_g^2 F(a)}{8\sqrt{2} qPE}\right)}{h^3 (E_g - E_i)},$$

$$F(a) = \frac{\pi}{2} + \sin^{-1}(1 - 2a) + 2(1 - 2a)\sqrt{a(1 - a)},$$

$$a = E_i/E_g,$$

to analyze, where m_e^* is the electron effective mass, P is the Kane matrix element (8.49×10^{-8} eV·cm [10]), E_g is the bandgap energy, M is the transition matrix element between the trap level and conduction band, N_i is the density of traps in the bulk, and all other symbols have their usual meaning. In narrow bandgap semiconductors, the tunneling current density J_{BBT} is highly sensitive to the local electric field E :

$$J_{BBT} \propto \exp\left(-\frac{\pi(m^*)^{1/2} E_g^{3/2}}{2qhE}\right)$$

where m^* is the effective mass of the charge carriers, E_g is the material bandgap energy, h is the reduced Planck constant, and E is the localized electric field magnitude. Extracted macroscopic current-voltage ($I-V$) evaluations confirm that by smoothing the field gradient, the optimal design avoids tunneling breakdown. As detailed in Fig. 4(f), the total

dark current of the optimal structure is remarkably well-controlled. Under low to moderate biases (< 5 V), the dark current is dominated by TAT and Shockley-Read-Hall (SRH) generation, maintaining an extremely low baseline near 10^{-14} A. Even as the avalanche mechanism begins to dominate at higher biases, the BBT component is strictly suppressed due to the sculpted electric field. At an operating bias of 10 V, where the device achieves an avalanche gain exceeding 100 (as seen in Fig. 4(e)), the total dark current is strictly confined to the 10^{-10} A level. Over a vast operational bias window, the macroscopic dark current density is strictly confined to the lowest possible background noise floor. Through this intricate, multidimensional balancing act, this research establishes a highly robust, secure operational parameter envelope for next-generation infrared sensors. Compared to recent state-of-the-art devices, our optimally sculpted architecture secures highly competitive performance while maintaining significantly lower manufacturing precision requirements and high process compatibility.

4. Conclusion

Driven by the rapidly escalating system-level demands for single-photon sensitivity, ultra-low dark current, and ultra-fast temporal resolution in modern infrared optoelectronics, this research has systematically deconstructed the microscopic physical mechanisms that govern macroscopic performance in HgCdTe Avalanche Photodiodes. Utilizing a highly advanced, first-principles microscopic-mesoscopic Monte Carlo theoretical framework, this study revealed that the macroscopic anomalies and bandwidth limitations plaguing legacy APD structures are fundamentally caused by unoptimized spatial aberrations in the internal electrostatic field.

This report introduced the conceptual framework of Electric Field Sculpting to explicitly demonstrate how microscopic doping differentials and specific geometric dimensions actively shape non-equilibrium carrier transport. It was established that extreme doping discrepancies between adjacent epitaxial regions carve steep electric field gradients.

These sharp transitions violently eject charge carriers into low-field zones immediately following impact ionization, inducing massive kinetic energy hemorrhaging via resonant polar optical phonon (POP) emission. This highly dissipative process shatters the deterministic spatial memory of the dead-space effect, leading directly to chaotic avalanche cascades, catastrophic gain fluctuations, and the severe inflation of the excess noise factor. Furthermore, arbitrarily increasing the i-region's physical width routinely forces the device into a state of incomplete depletion. The resulting formation of an electrically neutral dead zone forces carriers into sluggish diffusion, creating a time-domain diffusion tail that permanently throttles the device's high-frequency responsiveness to the megahertz regime.

Through rigorous, multidimensional physical evaluation, this study architected the ideal electrostatic profile. By seamlessly matching the functional region doping levels to eliminate steep gradients, and precisely aligning the physical i-region thickness with the maximum fundamental depletion limit, the optimal structure successfully achieves a deeply penetrating, gently sloping electric field. Microphysically, this meticulously sculpted landscape orchestrates a highly rhythmic, ultra-low-variance impact ionization cascade free of phonon drag. Macroscopically, it ensures total volumetric depletion, entirely excising the diffusion tail to unlock highly stable multi-GHz bandwidths. Furthermore, by inherently maintaining the peak field below the critical tunneling threshold, the structure effectively suppresses dark current while outputting signals at theoretical-limit low-noise and high avalanche gain. The complex cross-scale physical linkages elucidated in this comprehensive work provide a deeply predictive, highly actionable theoretical blueprint for the epitaxial design, structural optimization, and fabrication of the ultimate generation of high-speed, low-noise HgCdTe linear-mode avalanche focal plane arrays.

CRediT authorship contribution statement

Anna Liu: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Wei Gao:** Software, Formal analysis. **Hao Xie:** Data curation. **Chuan Shen:** Investigation. **Qing Li:** Visualization. **Hailu Wang:** Formal analysis. **Runzhang Xie:** Data curation, Conceptualization. **Lu Chen:** Methodology. **Piotr Martyniuk:** Investigation, Formal analysis. **Wen Lei:** Resources, Methodology. **Shuning Liu:** Writing – original draft, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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